Section (2) :

1. **Explain what is meant by (distribution) transparency, and give examples of different types of transparency.**

Transparency hides implementation details from users.

Four main types:

- Distribution transparency
- Transaction transparency
- Performance transparency
- DBMS transparency

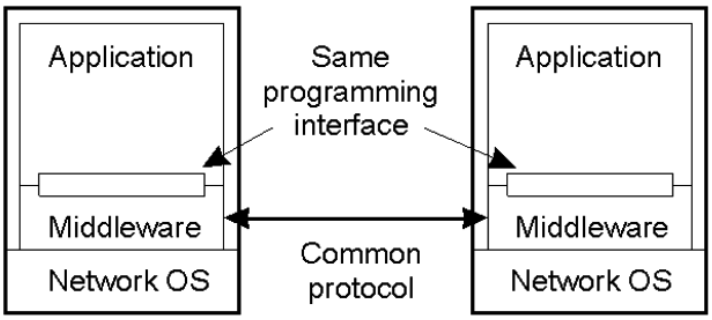
Distribution Transparency is the phenomenon by which distribution aspects in a system are hidden from users and applications.

Examples include access transparency, location transparency, migration transparency, relocation transparency, replication transparency, concurrency transparency, failure transparency, and persistence transparency.

2. **What is the role of the middleware in a distributed system?**

Middleware is a class of software technologies designed to help manage the complexity and heterogeneity inherent in distributed systems.

It is defined as a layer of software above the operating system but below the application program, that provides a common programming abstraction across a distributed system.

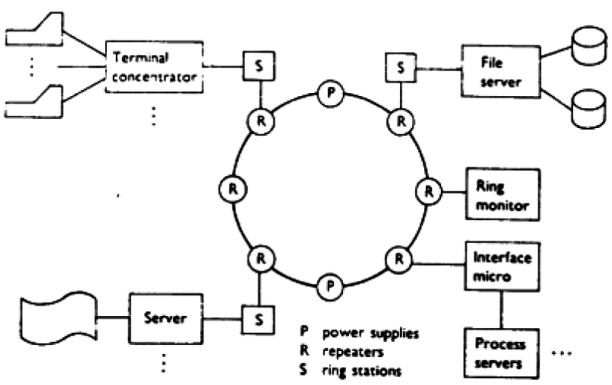


3. **Explain the main motivation for developing distributed computer systems?**

1. Development of PCs
2. Development of powerful microprocessors, microcomputers, and minicomputers
3. Development of high-speed networks
4. The need for multimedia for efficient decision making

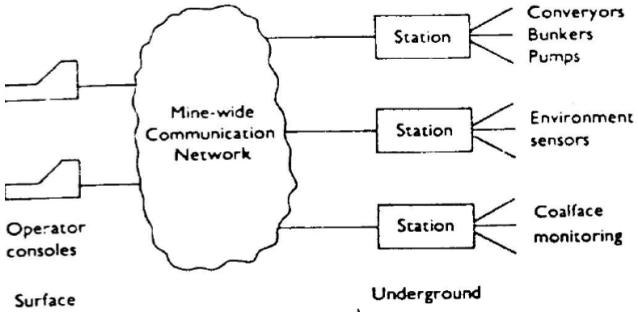
4. Draw and Explain Distributed Processing using Cambridge information system?

The objective was to provide a high performance system at low cost.
The only network device used was "Repeaters"
File Server —> " Data Storage "
Interface Micro —> " Hardware "
Ring Monitor —> " Software "
Process Server —> " Control "



5. Draw and explain the Mining control and monitoring distributed computer system?

The main characteristics of this system is that there are hundreds of PCs Distributed over 20 km and the system is real time.



6. Draw and Explain Distributed Processing using Bank of America information system?

Reliability:
What if the central computer goes down ?
- Standby PCs would be expensive.

Performance:
Would the responses for queries be sufficient ?
What about the future ?

Extensibility:
- New Larger central computers.

Maintenance:
For both hardware & software ?
- Insufficient redundancy of the system.

